Problem Set 4

1. Rotating space station

In order to create an Earth-like environment for its inhabitants, a station is rotated.

1. Find the angular speed required for the centripetal acceleration to equal g.
2. Express the period of rotation T in terms of R and g.

2. Circular motion of Earth

Earth is spinning about its axis with a period of 23 hours 56 min.

1. Find the velocity of a person at EPFL.
2. Find the person's centripetal acceleration.

3. Find the error: Sliding block

This looks like the solution to the problem, but it contains one or more errors.